

Yes, you have the core architecture down perfectly, but to be completely precise: **we have covered all the *major category types* of predictor sets, but we haven't listed out every single individual trace gas recipe.**

While the exact polynomial strings are proprietary mathematical optimizations, the full structural layout of SARTA's predictor inventory can be divided into five specific categories.

The complete breakdown of the distinct predictor sets SARTA runs under the hood includes:

1. The Fixed Gas Predictor Set (10 Terms)

- **The Inputs:** Uses a baseline profile where gases like CO₂, O₂, and N₂O are treated as a combined, uniformly mixed single gas density.
- **The Terms:** A 10-term polynomial mapping localized temperature deviations (ΔT) against the geometric view path ($\sec\theta$).

2. The Water Vapor Predictor Set (11 to 13 Terms)

Because water vapor absorbs across almost the entire infrared spectrum and exhibits self-broadening and foreign-broadening behavior, it uses a dedicated, highly complex expansion:

- **The Inputs:** Column density ratio (Q/Q_{ref}), layer temperature deviation (ΔT), and slant path ($\sec\theta$).
- **The Expansion:** Includes unique **square-root terms** ($\sqrt{Q/Q_{ref}}$) and **quadrant scaling** to linearize strong, saturated lines in the lower troposphere.

3. The Ozone Predictor Set (10 Terms)

Ozone has a highly unique vertical distribution that peaks in the stratosphere, requiring its own localized scaling scheme.

- **The Inputs:** Independent column mass ratio (V/V_{ref}) combined with standard path length and temperature metrics.

4. The Variable Trace Gas Predictor Sets (7 to 11 Terms per gas)

Depending on the version of SARTA being run (such as the standard operational NASA AIRS or NOAA CrIS versions), the algorithm initializes **separate independent predictor loops** for highly variable trace gases. These are evaluated individually:

- **Carbon Monoxide (CO) Predictor Set** (Crucial for shortwave infrared channels).
- **Methane (CH₄) Predictor Set** (Crucial for mid-wave infrared channels).
- **Nitrous Oxide (N₂O) Predictor Set** (Varies dynamically in specialized versions).
- **Nitric Acid (HNO₃) Predictor Set** (Often tracked for upper-troposphere/stratosphere chemistry).

5. The Cloud Scattering Predictor Set (Macro-Physical Parameters)

When the model switches out of "Clear Sky" mode, it activates a separate parameterization layer for two-stream radiative transfer calculations: [1]

- **The Inputs:** Cloud top pressure (P_{top}), cloud optical depth (τ), effective particle diameter (D_e), and fraction (C_f). These parameters are used to scan lookup tables (LUTs) rather than raw algebraic polynomials. [1]

Summary of the Flow

You accurately mapped the pipeline. SARTA runs all five of these predictor loops simultaneously for a given profile. It sums the resulting optical depths layer-by-layer, channel-by-channel, and feeds that final 100-layer optical grid into the vertical Schwarzschild integration scheme to deliver the final Top-of-Atmosphere brightness temperatures.

I can provide the **exact 11-term formula layout for the water vapor set** if you want to see how the non-linear math handles humidity, or we can discuss how SARTA addresses **atmospheric aerosols (like dust or volcanic ash)**. Which would you prefer to explore?

[1] <https://airs.jpl.nasa.gov>